

# Synthesise. Then judge.

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An open challenge — pick a dataset, generate, and your own utility and privacy evaluations.

 **Sam Urmian** · [Slate](#)

4 datasets · TabDM + any LLM · ~60 minutes





# Your task

Generate a synthetic copy with **TabDM** and a synthetic with **any LLM**. design your own metrics for **utility** and **privacy** — and judge which wins on each.



## Use any tools

TabDM is required. The LLM is your choice — OpenAI, local, anything.



## Invent the metrics

No prescribed evaluation. Decide what 'useful' and 'private' mean for your dataset.



## Share the verdict

Be ready to defend your method to the room in the wrap-up.



# The four datasets

All ~2,000 rows, all pre-split (1,399 train · 300 valid · 300 test).



higher\_ed\_dropout\_success.csv

## Higher-Ed Dropout & Success

**39 columns** · 3-class · dropout / enrolled / graduate

Graduate 998 · Dropout 642 · Enrolled 359

*Demographics + per-semester course outcomes — rich, mixed types.*



oulad\_final\_result.csv

## OULAD Final Result

**14 columns** · 4-class · pass / fail / withdrawn / distinction

Pass 757 · Withdrawn 623 · Fail 433 · Distinction 186

*Compact OULAD-style record with region, IMD band, age band.*



student\_performance\_pass\_fail.csv

## Student Performance (Math /

**37 columns** · Binary · pass / fail

Pass 1559 · Fail 440

*Survey-style features — family, study habits, grades G1/G2/G3.*



xapi\_edu\_performance\_level.csv

## xAPI Educational Performance Level

**19 columns** · 3-class · L / M / H performance level

Medium 879 · High 591 · Low 529

*Behavioural — raised hands, visited resources, parent involvement.*



# Utility — design your own measure

*There is no official answer. What does "useful" mean for your dataset?*



## UTILITY

*Does this synthetic data  
let me do something useful?*



### Could you train a model on it?

TSTR-style: train on synthetic, test on real. F1, AUC, anything fits your task.



### Would the descriptive stats fool a colleague?

Compare means, distributions, correlations against the real train set.



### Does it preserve what matters most to YOU?

Pick the one feature or relationship your research depends on. Check it.



### Could a student learn from it?

Eyeball ten random rows. Do they look like plausible students?



# Privacy — design your own check

*"It's synthetic" is not a guarantee. Decide what would convince you the synthetic data is safe.*



## PRIVACY

*Could a real student  
be identified from this?*



### Did the model memorise?

How many synthetic rows are exact duplicates of training rows? Anything  $> 0$ ?



### How close is too close?

For each synthetic row, what's the distance to its nearest real neighbour?



### Could you re-identify a rare student?

Pick someone unusual in the real data. Can you find them in the synthetic copy?



### What would a regulator ask?

If your IRB saw this, would they accept it as a substitute for the real data?



# How the hour runs

*Pair up if you'd like — comparing methods is half the value.*



## Pick & explore

Choose a dataset. Look at it. What's the distribution of the target? What features look weird?



## Generate

Run TabDM (pip install tabdm). Run any LLM with your favourite prompt. Save both synthetic copies.



## Evaluate

Build the utility check and the privacy check that you'd actually trust. No prescribed metrics.



## Share

One slide / one paragraph / one chart — defend your verdict to the room.



# Share-back — defend your verdict

*Last 10 minutes. Tell us what you measured, why, and which generator won.*

**Q1** Which dataset did you pick, and what made you pick it?

**Q2** What single utility metric did you trust most — and why?

**Q3** What single privacy check did you trust most — and why?

**Q4** TabDM or LLM: which won on utility? Which won on privacy? Were they the same?

**Q5** What did your evaluation MISS? If you had a week, what would you build next?

# Now go synthesise something useful.

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The fastest way to learn synthetic data is to break some.  
Pick the dataset closest to your work and start there.



## Code

[github.com/mlgorithm/tabdm](https://github.com/mlgorithm/tabdm)



## Materials

[lasd.slaterresearch.ai](https://lasd.slaterresearch.ai)



## Ask

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